

Question 1

create the **class Course** with the below Attributes:

courseId- int

courseName- String

courseAdmin- String

quiz- int

handson -int

*The above **methods should be private** ,write getter and setter and **parametrized constructor** as required.*

create **class courseProgram** with **main method**.

implement **two static methods-**

findAvgOfQuizByAdmin method:

- this method will take array of Course objects
- and a String value as input parameters.

This method will find out Average (as int) of Quiz questions
for given Course Admin (String parametre passed)

This method will return Average if found.

- if there is no course
that matches then the method **should return 0**.

sortCourseByHandsOn method:

- This method will take an Array of Course Objects
- and int value as input parameters.

This methods should ***return an Array of Course objects in an ascending order of their handson which are less than the given handson(int parameter passed) value.***

- ✓ if there is no such course then the method should return null.

The above mentioned static methods should be called from main methods.

for *findAvgOfQuizByAdmin method*:

- The main method should print the average if the returned value is not 0.
- if the returned value is 0 then it should print "No Course found."

for *sortCourseByHandsOn method*:

- the main method should print the name of the Course from the returned Course object Array if the returned value is not null.
- if the returned value is null then it should print "No Course found with mentioned attribute."

input1:

111

kubernetes

Nisha

40

10

321

cassandra

Roshini

30

15

457

Apache Spark

Nisha

30

12

987

site core

Tirth

50

20

Nisha

17

output1:

35

kubernetes

Apache Spark

cassandra

input2:

111

kubernetes

Nisha

40

10

321

cassandra

Roshini

30

15

457

Apache Spark

Nisha

30

12

987

site core

Tirth

50

20

Shubhamk

5

output 2:

No Course found

No Course found with mentioned attributes.

Question 2

create a class **Footwear** which consists of the below attributes.

footwearId=int

footwearName=String

footwearType=String

price =int

the above attributes should be private.

write getter and setter and parametrised constructor as required.

create the class footwearProgrammm with the main method.

*implement the 2 static methods.**getCountByType** and*

getSecondHighestPriceByBrand in the Solution class.



getCountByType method:

this method will take two input parameters.

- array of the Footwear objects and string parameter footwear type.
- this method will return the count of the footwears from array of the footwear objects for the given type of footwear.
- if no footwear with the given footwear type is found in the array of footwear objects, then the method should return 0.

getSecondHighestPriceByBrand method:

this method will take 2 input parameters-

- array of footwear objects and string parameter inputFootwearName. the method
- will return the **second highest footwear objects** based on **the price from the array of the Footwear objects** whose **brand name** matches with the **input string parameter.**
- if no footwear with the given brand is present in the array of the footwear objects, then the method should **return null.**

NOTE: no two footwear objects would have the same footwearId. All the searches should be case insensitive.

the above mentioned static methods should be called from the main method.

- ❖ **for getCountByType method-** the main method should print the count of the footwears, if the returned value is greater than zero. or it should print "Footwear not available";

❖ for **getSecondHighestPriceByBrand** method-The main method should print price from the returned footwear objects

❖ if the returned footwear object is not null.else it should print "Brand not available".

for example.

112

ABC

25555

where 112 is the footwear id,ABC is brand name,25555 is price.

consider the sample input and output.

100

Sketchers

sneakers

12345

103

Puma

running shoes

10099

102

reebok

Running shoes

5667

101

Reebok

running shoes

5656

99

reebok

floaters

5666

Running shoes

reebok

Sample output:

3

99

reebok

5666

Sample input2:

100

Puma

sneakers

12345

101

Puma

sneakers

10099

102

Puma

sneakers

5000

102

Reebok

sneakers

8000

104

Puma

floaters

2000

running shoes

bata

Sample output:

Footwear not available

Brand not available

Question 3

Create a **class called Student** with the below attributes:

rollNo - int

name - String

branch - String

score - double

dayScholar - boolean

The above **attributes** should be **private**, write **getters**, **setters** and **parameterized constructor** as required.

✓ Create **class Solution** with **main method**.

Implement two static methods -**findCountOfDayscholarStudents** and **findStudentwithSecondHighestScore** in **Solution** class.

findCountOfDayscholarStudents:

- ✓
- ***This method will take an array of Student objects as an input parameter .***
 - This method will calculate and return the count of Students whose score is greater than 80 and who are all from dayScholar.
 - If no Student scored greater than 80 and from dayScholar are present in the array of Student objects, then the method should return 0.

findStudentwithSecondHighestScore:

- This method will take an array of Student objects as an input parameter.
- This method will return the object of **the second highest score student from the array of Student objects** who are not from the dayScholar.
- If no Student is a dayScholar in the array of Student objects, then the method should return null.

Note : All the searches should be case insensitive.

- ❖ The combination of dayScholar and score for each student is always unique.
- ❖ The above mentioned static methods should be called from the main method.

For findCountOfDayscholarStudents method - The main method should print the returned count as it is if the returned value is greater than 0, else it should print "**There** are no such dayscholar students".

For findStudentwithSecondHighestScore method - The main method should print the rollNo, name and score in the below format from the returned object if the returned value is not null.

rollNo#name#score

- ✚ If the returned value is null, then it should print "There are no student from non day scholar"

Before calling these static methods in main, use Scanner object to read the values of four Student objects referring attributes in the above mentioned attribute sequence.

Consider below sample input and output:

Input:

1001

Ashwa

IT

85

true

1002

Preeti

IT

70

false

1003

Uma

ECE

85

false

1004

Akash

EEE

90

True

Output:

2

1002#Preeti#70.0

Question 4

Create a **class College** with the below attributes.

id-int

name -String

contactNo-int

address-String

pinCode-int

 Write the getters and setters and parametrized constructor in the above mentioned sequence as required.

Create the class **Solution** with the main method.

Implement the two static methods:

1.findCollegeWithMaximumPincode

2.searchCollegeByAddress

 **findCollegeWithMaximumPincode method:**

 Create the Static method in the Solution Class.

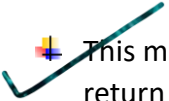
 This method will take array of the College objects and return the College object having maximum pincode if found else return null if not found.

 for this method ,main method will print College object with maximum pincode if the returned value is not null.

 if the returned value is null ,then the main method will print "No college found with mentioned attribute".

 searchCollegeByAddress **method:**

 Create the Static method in the Solution Class.

 This method will take array of College objects as input and address as input and return College object having the mentioned address if found else return null if not found.

 for this method main method will print College object details as it is ,

 if the returned value is not null.if the returned value is null then ,main method will print "No college found with mentioned attribute".

input 1:

4

109

ACT

2500256

mumbai

695001

107

MCE

2500254

malapuram

612354

113

CTE

2500252

chennai

623145

102

SCT

2500255

AP

523641

AP

OUTPUT1:

id-109

name-ACT

contactNo-2500256

address-mumbai

pincode-695001

id-102

name-SCT

contactNo-2500255

address-AP

pincode-523641

INPUT2:

4

111

MJT

2500251

Calicut

401235

105

MET

2500256

kochi

668745

115

IIT

2500262

banglore

569874

110

ACT

2500263

delhi

687945

delhi

output2:

id-110

name-ACT

contactNo-2500263

address-delhi

pincode-687945

id-110

name-ACT

contactNo-2500263

address-delhi

pincode-687945

QUESTION 5

Create a class **Motel** with the below attributes:

motelId - int

motelName - String

dateOfBooking – String (in the format dd-mon-yyyy)

noOfRoomsBooked – int

cabFacility – String

totalBill- double

- The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class **Solution** with main method.

Implement one static method – **totalNoOfRoomsBooked** in **Solution** class.

totalNoOfRoomsBooked method:

- This method will take two input parameter - **array of Motel objects** and a **String parameter**.
- The method will return the **total numbers of rooms** booked from array of Motel objects if the **cab facility** attribute matches with **the given String parameter(cab facility)** and **the number of rooms booked is greater than 5**.
- If no rooms are booked with the above criteria in the array of Motel objects, then the method should **return 0**.

Note :

- No two Motel object would have the same motelId.
dateOfBooking is stored in the format dd-mon-yyyy(eg. 01-Jan-2022)

The above mentioned static method should be called from the main method.

For totalNoOfRoomsBooked method –

- The main method should print the total number of booked rooms as it is, if the returned value is greater than 0, else it should print **"No such rooms booked"**
- Before calling these static methods in main, use Scanner object to read the values of four Motel objects referring attributes in the above mentioned attribute sequence.
- Next, read the value of one String parameter for capturing the cab facility

Input

1001

M&M

01-Dec-2022

5

Yes

30000

1002

BestStay

10-Jan-2022

3

Yes

27000

1003

Novatel

11-Jun-2022

5

Yes

25000

1004

Chola

01-Sep-2022

7

Yes

72000

Yes

OutPut

7

Question 6

Create the **class RRT(Rapid response team)** with the below attributes:

ticketNo-int

raisedBy-String

assignedTo-String

priority-int

project-String

All attributes should be private,write getters and setters and

parameterized constructor as required.

Create class MyClass with main method.

Implement a static method-getHighestPriorityTicket in MyClass class.

getHighestPriorityTicket method:

- This method will **take an array of RRT objects ,and a String value as parameters.**
- This method will **return the RRT object** with highest priority ticket from the array of the RRT objects for the **given project(String parameter passed).** Highest priority is the one which has lesser value.

for example: 1 is considered as high priority and 5 is considered as low priority.

- If no RRT with the above condition is present in the array of the RRT objects, then the method should return null.

The main method should **print the ticketNo, raisedBy and assignedTo** from returned object **if the returned object is not null.** if the **returned object is null** then **main method should print "No such Ticket".**

input1:

123

Velantish

Mani

3

Xperience

234

Sathish

Akshaya

1

AIG

345

John

Jack

2

AIG

456

Bhuvi

Jack

5

AIG

AIG

output1:

234

Sathish

Akshaya

Input2:

123

Velantish

Mani

3

Xperience

234

Sathish

Akshaya

1

AIG

345

John

Jack

2

AIG

456

Bhuvi

Jack

5

AIG

Xplore

output2:

No such ticket.

Question no 7

Create a **class Sim** with below attributes:

simId - int

customerName - String

balance - double

ratePerSecond - double

circle - String

Write getters, setters and parameterized constructor as required.

Public class Solution is already created with main method.

Code inside main method should not be altered else your solution might be scored as zero. You may copy the code from main method in eclipse to verify your implementation.

- Implement static method - **transferCustomerCircle** in Solution class.
- This method will take **first parameter** as array of Sim class objects,

- **second parameter as circle to be transferred** (which is String parameter circle1) and **third parameter as new circle** (which is String parameter circle2)

❖ Method will transfer **the customer to new circle (circle2)**, where the

circle attribute would match **second parameter (circle1)**.

🚦 Method will return array of Sim objects for which circle is transferred.

Return array should be sorted in descending order of ratePerSecond

(assuming ratePerSecond is not same for any of the Sim objects).

This method should be called from main method and display the simId,

customerName,circle and ratePerSecond of returned objects

(as per sample output).

Main method mentioned above already has Scanner code to read values,

create objects and test above methods. Hence do not modify it.

Consider below sample input and output:

Input:

1

raj

100

1.5

KOL

2

chetan

200

1.6

AHD

3

asha

150

1.7

MUM

4

kiran

50

2.2

AHD

5

vijay

130

1.8

AHD

AHD

KOL

Output:

4 kiran KOL 2.2

5 vijay KOL 1.8

2 chetan KOL 1.6

Question 8

Create a **class Hotel** with the below attributes:

hotelId - int

hotelName - String

dateOfBooking – String (in the format dd-mon-yyyy)

noOfRoomsBooked – int

wifiFacility – String

totalBill- double

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create **class Solution with main method.**

Implement two static methods – noOfRoomsBookedInGivenMonth and searchHotelByWifiOption in Solution class.

noOfRoomsBookedInGivenMonth method:

This method will **take two input parameter** - array of Hotel objects and a String parameter.

The method will return **the total numbers of rooms booked from array of Hotel objects for the given month**(String parameter passed).

If no rooms are booked for the given month in the array of Hotel objects, then the method should return 0.(**This is repeated pattern**)--later

searchHotelByWifiOption method:

This method will **take two input parameter** - array of Hotel objects and a String parameter

The method will return **Hotel object** with **second highest totalbill**, from the array of Hotel objects where **wifiFacility attribute matches with the given wifi facility**(String parameter passed).

If no Hotel with the given wifi option is present in the array of Hotel objects, then the method should return null.

Note :

- ❖ No two Hotel object would have the same hotelId.
- ❖ No two Hotel object would have the same totalbill.
- ❖ The Array either has at least two objects with specified wifi option OR no object with specified wifi option.
- ❖ All the searches should be case insensitive.
- ❖ **dateOfBooking is stored in the format dd-mon-yyyy(eg. 01-Jan-2022)**

The above mentioned static methods should be called from the main method.

For noOfRoomsBookedInGivenMonth method - The main method should print the total number of booked rooms as it is,

if the returned value is greater than 0, else it should print "No rooms booked in the given month"

For searchHotelByWifiOption method - The main method should print the hotelId from the returned Hotel object if the returned value is not null.

If the returned value is null then it should print "No such option available".

Before calling these static methods in main, use Scanner object to read the values of four Hotel objects referring attributes in the above mentioned attribute sequence.

Next, read the value of two String parameters for capturing the month and wifi option .

Consider below sample input and output:

Input1:

101

Best Stay

01-jan-2022

10

Yes

20000

102

Apple Stay

12-Feb-2022

3

Yes

4000

103

Accord

11-May-2022

5

Yes

15000

104

Royal Park

22-Dec-2021

7

Yes

12000

May

Yes

Output1:

5

103

Input2:

101

Best Stay

01-jan-2022

10

Yes

20000

102

Apple Stay

12-Feb-2022

3

Yes

4000

103

Accord

11-May-2022

5

Yes

15000

104

Royal Park

22-Dec-2021

7

Yes

12000

May1

Yes1

Output2:

No rooms booked in the given month

No such option available

Sample code snippet for reference:

Please use below code to build your Solution.

```
import java.util.Scanner;

public class Solution
{
    public static void main(String[] args)
    {
        //code to read values
```

```
//code to call required method  
//code to display the result  
}
```

```
//code the first method
```

```
//code the second method
```

```
}
```

```
//code the class
```

Note on using Scanner object:

Sometimes scanner does not read the new line character while invoking methods like `nextInt()`, `nextDouble()` etc.

Usually, this is not an issue, but this may be visible while calling `nextLine()` immediately after those methods.

Consider below input values:

1001

Savings

Referring below code:

```
Scanner sc = new Scanner(System.in);
```

```
int x = sc.nextInt();
```

```
String str = sc.nextLine(); -&gt; here we expect str to have value Savings. Instead it may be "".
```

If above issue is observed, then it is suggested to add one more explicit call to `nextLine()` after reading numeric value.

Question 9

Create a **class Book** with below attributes

id - int

pages - int

title - String

author - String

price - double

The above attributes should be private, write getters and setters and parameterized constructor as required.

Create a class **Solution** with main method

Implement two static methods - `findBookWithMaximumPrice` and `searchBookByTitle` in **Solution** class.



`findBookWithMaximumPrice :`

=====

- Create a static **method findBookWithMaximumPrice** in the **Solution** class. This method will take **array of Book objects** and
- returns **the Book object having the maximum Price** if found else return null if not found.

searchBookByTitle :



=====

- Create a static method **searchBookByTitle** in the Solution class. This method will take array of **Book objects** and **Title as input** and returns the Book object having the mentioned Title if found else return null if not found.(Easy just comparison)
- These methods should be called from the main method.

write code to perform the following tasks.

1. Take necessary input variable and call findBookWithMaximumPrice. For this method - The main method should print the Book

object with the maximum of mentioned attribute as it is if the returned value is not null, or it should print

"No Book found with mentioned attribute."

2. Take necessary input variable and call searchBookByTitle. For this method - The main method should print the Book object

details as it is, if the returned value is not null or it should print "No Book found with mentioned attribute."

The above mentioned static methods should be called from the main method. Also write the code for accepting the inputs and

printing the outputs. Don't use any static test or formatting for printing the result. Just invoke the method and print the

result.

All String comparison needs to be in case sensitive.

Input:

1

845

Bengali

Arijit

525.50

2

456

English

Raju

412.30

3

1022

History

Kaka

525.50

4

125

geography

MN

524

English

Output:

1 Bengali

3 History

2

456

QUESTION 10

Create a class called **Employee** with the below attributes:

employeeId - int



name - String

branch - String

rating - double

company Transport - boolean

The above attributes should be private. write getters, setters and parameterized constructor as required.

Create class MyClass with main method.

Implement two static methods **findCountOfEmployeesUsingCompTransport** and **findEmployeeWithSecondHighestRating** in MyClass class.

findCountOfEmployeesUsingCompTransport method:

- This method will take **an array of Employee object** and a **String parameter** as input parameters.
- This method will calculate and **return the count of Employees** who are all **using company transport in the given branch (String parameter passed)**.
- If no Employee in the given branch using company transport in the array of Employee objects, then
- the method should return 0.

findEmployeeWithSecondHighestRating method:

- This method will take **an array of Employee objects** as an input parameter. This method will return
- the **object of the second highest rating** employee from the array of Employee objects who are not using
- company transport.
- If all Employees using company transport in the array of Employee objects, then the method should return null.
- Note: All the searches **should be case sensitive**, companyTransport and rating combination of each **Employee is unique**.

For **findCountOfEmployeesUsingCompTransport** method, The **main method should print** the **returned count as it is** if the **returned value is greater than 0**, else it should print **"No such Employees"**.

Ex: 2 , where 2 is the count

For findEmployeeWithSecondHighestRating method: The main method should print the employeeid and name from

the returned object if the returned value is not null. If the returned value is null, then it should print "All Employees using company transport".

Ex: 1003

Uma

where 1003 is the employeeid and Uma is the name.

Before calling these static methods in main, use Scanner object to read the values of four Employee objects referring attributes in the above mentioned attributes sequence. Next, read the value of String parameter for capturing branch.

Input :

1001

Ashwa

IT

5

true

1002

Preeti

IT

4

true

1003

Uma

Admin

3

false

1004

Akash

Hardware

4.5

false

IT

Output :

2

1003

Uma

Question 11

Create a class Player with below attributes:

playerId - int

skill - String

level - String

points - int

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create class Solution with the main method.

Implement two static methods - findPointsForGivenSkill and getPlayerBasedOnLevel in Solution class.

findPointsForGivenSkill method:

This method will take two input parameters - array of Player objects

and string parameter skill. The method will return the sum of the points attribute from player objects for the skill passed as parameter. If no player with the given skill is present in the array of player objects, then the method should return 0.

getPlayerBasedOnLevel method:

This method will take two String parameters level and skill, along with the array of Player objects. The method will return the player object, if the input String parameters matches with the level and skill attribute of the player object and its point attribute is greater than or equal to 20. If any of the conditions are not met, then the method should return null.

Note : No two player object would have the same value for skill and level attribute. All player object would have the points greater than 0. All the searches should be case insensitive.

These above mentioned static methods should be called from the main method.

For findPointsForGivenSkill method - The main method should print the points as it is if the returned price is greater than 0, or it should print "The given Skill is not available".

For getPlayerBasedOnLevel method - The main method should print the playerId of the returned player object.

If the returned value is null then it should print "No player is available with specified level, skill and eligibility points".

Before calling these static methods in main, use Scanner object to read the values of four Player objects

referring attributes in the above mentioned attribute sequence. Next, read the value for skill and level.

Please consider the skill value read above as skill parameter for both the static methods.

Input:

101

Cricket

Basic

20

102

Cricket

Intermediate

25

111

Football

Intermediate

50

113

BaseBall

Advanced

100

Cricket

Intermediate

Output:

45

102

QUESTION 11

Create a **class Player** with below attributes:

playerId - int

skill - String

level - String

points - int

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create class Solution with the main method.

Implement **two static methods - findPointsForGivenSkill and getPlayerBasedOnLevel** in Solution class.

findPointsForGivenSkill method:

- This method will **take two input parameters - array of Player objects and string parameter skill.**
- The method will return the **sum of the points attribute from player objects** for the skill passed as parameter.
- If no player with the given skill is present in the array of player objects, then the method should return 0.

getPlayerBasedOnLevel method:

- This method will **take two String parameters level and skill**, along with the **array of Player objects.**
- The method will **return the player object**, if the **input String parameters matches with the level and skill attribute of the player object** and its **point attribute is greater than or equal to 20**. If any of the conditions are not met, then the method should return null.

Note : **No two player object would have the same value for skill and level attribute.**

All player object would have the points greater than 0.

All the searches should be case insensitive.

These above mentioned static methods should be called from the main method.

For findPointsForGivenSkill method - The main method **should print the points as it is** if the returned price is greater than 0, or it should print **"The given Skill is not available"**.

For getPlayerBasedOnLevel method - **The main method should print the playerId of the returned player object.** If the returned value is null then it should print "No player is available with specified level, skill and eligibility points".

Before calling these static methods **in main, use Scanner object to read the values of four Player objects** referring attributes in the above mentioned attribute sequence. Next, **read the value for skill and level.** Please consider the **skill value read above as skill parameter for both the static methods.**

Input:

101

Cricket

Basic

20

102

Cricket

Intermediate

25

111

Football

Intermediate

50

113

BaseBall

Advanced

100

Cricket

Intermediate

Output:

45

102

Question 12

Create **class Medicine** with below attributes:

MedicineName - String

batch - String

disease - String

price - int

Create class Solution and **implement static method "getPriceByDisease" in the Solution class.**

- This method will take **array of Medicine objects** and a **disease String as parameters**. And **will return another sorted array of Integer objects** where the **disease String matches with the original array of Medicine object's disease attribute** (**case insensitive search**).

Write necessary getters and setters.

Before calling "getPriceByDisease" method in the main method, read values for four Medicine objects

referring the attributes in above sequence along with a String disease. Then call the "getPriceByDisease"

method and print the result.

Input

dolo650

FAC124W

fever

100

paracetamol

PAC545B

bodypain

150

almox

ALM747S

fever

200

aspirin

ASP849Q

flu

250

fever

Output

100

200

Question 13

Create a **class AutonomousCar** with the below attributes:

carId – int

brand – String

noOfTestsConducted – int

noOfTestsPassed- int

environment – String

Write getters, setters for the above attributes . **Create constructor which takes parameter in the above sequence except grade.**

Create class Solution with main method. Implement two static methods – findTestPassedByEnv and updateCarGrade in

Solution class.

findTestPassedByEnv method:

- ❖ This method will take **two input parameters -array of AutonomousCar objects and string parameter environment**. The method
- ❖ will return the **sum of the noOfTestsPassed attribute** from autonomousCar objects for the **environment passed as parameter**.
- ❖ If no autonomousCar with the given environment is present in the array of AutonomousCar objects, then the method should
- ❖ return 0.

updateCarGrade method:

- ❖ This method will take a **String parameter brand**, along with the **array of AutonomousCar objects**. The method will **return the autonomousCar object**, if the **input String parameter matches with the brand** attribute of the autonomousCar object.
- ❖ Before **returning the object, the grade should be derived based** on the **rating calculation** mentioned below.
- ❖ **This grade -**
 - **value should be assigned to the object**. If any of the above conditions are not met, then the method should return null.
- ❖ The **grade attribute** should be calculated as follows: **rating .(noOfTestsPassed * 100)/noOfTestsConducted**
- ❖ If the **rating > = 80 then grade should be 'A1'**, otherwise the **grade should be 'B2'**.

The above mentioned static methods should be called from the main method. For findTestPassedByEnv method – The main method should **print the totalTestPassed** as it is, if **the returned value is greater than 0, or it should print "There are no tests passed in this particular environment"**.

For updateCarGrade method – The main method should **print the brand and grade** of the returned **autonomousCar object**.

The **brand and grade should be concatenated with :: while printing**. eg:- Tesla::A1, where Tesla is the brand and A1

is the grade.

If the returned value is null then it should print “No Car is available with the specified brand”. Before calling these

static methods in main, use Scanner object to read the values of four autonomousCar objects referring attributes in the

above mentioned attribute sequence (except grade attribute). Next, read the value for environment and brand.

Input

100

Tesla

1000

500

Hills

200

Ford

2000

1500

Desert

300

Royce

3000

1700

Hills

400

Mercedez

1000

400

Desert

Desert

Mercedez

Output

1900

QUESTION 14

Create **class Movie** with below attributes:

movieName - String

company - String

genre - String

budget - int

Create class Solution and implement static **method "getMovieByGenre" in the Solution class.**

- ❖ This method will take array of Movie objects and a searchGenre String as parameters.
- ❖ And will return **another array of Movie objects** where the searchGenre String matches with the original array of Movie object's genre attribute (case insensitive search).

Write necessary getters and setters.

Before calling "getMovieByGenre" method in the main method, read values for four Movie objects referring the

attributes in above sequence along with a String searchGenre. **Then call the "getMovieByGenre" method and write**

logic in main method to print "High Budget Movie",if the movie budget attribute is greater than 80000000 else

print "Low Budget Movie".

Input

aaa

Marvel

Action

250000000

bbb

Marvel

Comedy

25000000

ccc

Marvel

Comedy

2000000

ddd

Marvel

Action

300000000

Action

Output

High Budget Movie

High Budget Movie

QUESTION 15

Create a **class Phone** with below attributes:

phoneId - int

os - String

brand - String

price - int

Write getters, setters and **parameterized constructor** in the above mentioned attribute sequence as required.

Create class Solution with main method.

Implement two static methods - findPriceForGivenBrand and getPhoneIdBasedOnOs in Solution class.

findPriceForGivenBrand method:

- ❖ This method will take **two input parameters - array of Phone objects and string parameter brand.**
- ❖ The method will return the **sum of the price attribute from phone objects** for the brand passed as parameter.
- ❖ If no phones with the given brand is present in the array of phone objects, then the method should
- ❖ return 0.

getPhoneIdBasedOnOs method:

- ❖ This method will take a **String parameter os**, along with the **array of Phone objects.**
- ❖ The method will **return the phone object**, if the input String parameter matches with the os attribute of the phone object and **its price attribute is greater than or equal to 50000**. If any of the conditions are not met, then the method should return null.

Note : No phone object would have the same value for os attribute. All phone object would have the price greater than 0.

All the searches should be case insensitive.

These above mentioned static methods should be called from the main method.

For findPriceForGivenBrand method - The main method should print the price as it is if the returned price is greater

than 0, or it should print "The given Brand is not available".

For getPhoneIdBasedOnOs method - The main method should print the phoneId of the returned phone object. If the returned

value is null then it should print "No phones are available with specified os and price range".

Before calling these static methods in main, use Scanner object to read the values of four Phone objects referring

attributes in the above mentioned attribute sequence. Next, read the value for brand and os.

Input

111

iOS

Apple

30000

222

android

Samsung

50000

333

Symbian

HTC

12000

444

Paranoid

HTC

89000

Blackberry

aNdROID

Output

The given Brand is not available

222

QUESTION 16

Create a class **NavalVessel** with the below attributes:

vesselId – int

vesselName – String

noOfVoyagesPlanned – int

noOfVoyagesCompleted – int

purpose – String

Write getters, setters for the above attributes. **Create constructor which takes parameter in the above**

sequence except classification.

Create class Solution with main method. Implement two static methods – findAvgVoyagesByPct and findVesselByGrade

in Solution class.

findAvgVoyagesByPct method:

This method will take **two input parameters** -array of **NavalVessel** objects and **int parameter percentage**. The method will return **the average of the noOfVoyagesCompleted attribute from NavalVessel objects which meets the percentage requirement as mentioned below.**

The **percentage for each navalVessel** is calculated as

percentage $.(noOfVoyagesCompleted * 100)/noOfVoyagesPlanned$.

If the **above calculated percentage is greater than or equal to the percentage passed as the parameter**, then **those objects should participate in the average calculation**. If there are no objects with the mentioned criteria, then the method should return 0.

findVesselByGrade method:

- ❖ This method will take a **String** parameter purpose, along with **the array of NavalVessel objects**. The **method will return** the **NavalVessel object**, if the **input String parameter** matches with the **purpose attribute of the NavalVessel object**.
- ❖ Before returning the object, the **classification should be derived as mentioned below**. This **classification value should be assigned to the object before returning**. If any of the above conditions are not met, then the method should return null. The
- ❖ **classification attribute should be calculated based on the percentage calculation mentioned above.**
- ❖ If the **percentage = 100** then **classification should be 'Star'**, if percentage is **between 99 and 80** then classification should
- ❖ be **'Leader'**, if percentage is between **79 and 55** then classification should be **'Inspire'** **otherwise** the classification should
- ❖ be **'Striver'** If the returned value is null then it should print "No Naval Vessel is available with the specified purpose".

Before calling these static methods in main, use Scanner object to read the values of four NavalVessel objects referring

attributes in the above mentioned attribute sequence (except the classification attribute). Next, read the value for

percentage and purpose.

Input

111

Rojer

100

100

Passenger

222

Kalam

200

100

Goods

333

Yashwin

400

300

Army

444

Thanishwini

500

500

Welfare

75

Army

Output

300

Yashwin%Inspirer

QUESTION 17

Create a **class Student** with below attributes:

rollNo - int

name - String

subject - String

grade - char

date - String [DD/MM/YYYY]

The above attributes should be private. Write Getter and Setter and **parametrized constructor** as required.

Create class Solution with main method.

Implement **one static method: findStudentByGradeAndMonth** Solution Class.

findStudentByGradeAndMonth Method:

This method will take an array of **Student objects**, **char value as grade and int value as month** for input parameters.

The method will **find out all Students from the given grade and month**.

This method will **return array of Student object** **ascending based** on **their rollNo** if found.

If there is no Student that matches then the method should return null.

for this method- **main method should print Student name, subject and total student** found [The length of the list], if the returned value is not null. If the returned value is null then main method should print "No student found".

NOTE:

1. For Taking char as input use `sc.nextLine().charAt(0)`
2. To match/check the month You **havee to convert int month in the parameter to String.**
3. No need to **count the Student array if return not null just print the array length.**

Consider the below input and output:

input1:

111

Arijit

Math

B

22/09/2023

101

Priyanka

English

A

30/03/2022

107

Shreosi

History

C

13/05/2022

105

Tatan

Physics

A

27/03/2022

A

3

output1:

Priyanka

English

Tatan

Physics

2

input2:

111

Sohel

Math

B

22/09/2022

101

Priyanka

English

A

30/03/2022

107

Gopa

History

C

12/05/2022

105

Kamal

Physics

A

27/03/2022

A

7

output 2:

No student found

QUESTION 18

create a class **Antenna** with below attributes.

antennaId=int

antennaName=String

projectLead=String

antennaVSWR=double

the above attribute should be private ,write getter and setter and **parametrized constructor** as well.

create class MyClass with main method.

implement two static methods searchAntennaByName and sortAntennaByVSWR in MyClass class.

searchAntennaByName :

- ❖ This method will take **an array of Antenna objects** and the **string value as input** parameter.
- ❖ The method will find **out Antenna name(String parameterpassed)**.
- ❖ It will return **Antennaid** if found.
- ❖ if there is no Antenna that matches then the method will return zero.
- ❖ The main method **should print the antennaid**,if the returned value is not 0.if the returned value is 0 then print,"There is no antenna with the given parameter".

sortAntennaByVSWR:

This method will take an **array of Antenna Objects** and a **double value as input**.

This method will **return an array of Antenna objects** sorted in **ascending orderof their antennaVSWR**, which **is less than VSWR(double value passed)**.

The main method should **print name of the project leads from the returned array** if the **returned value is not null**.if the returned value is null then main method

should print "No Antenna found";

input1:

111

Reconfigurable

Hema

0.4

222

Wearable

Kavya

0.9

333

Microstrip

Teju

0.3

444

Dielectric

Sai

0.65

Microstrip

0.5

output:

333

Teju

Hema

input 2:

111

Reconfigurable

Hema

0.3

222

Wearable

Kavya

0.9

333

Microstrip

Teju

0.4

444

Dielectric

Sai

0.65

Resonator

0.25

output:

There is no antenna with given parameter

No Antenna found

QUESTION 19

Create the **class Flowers** with below attribute:

flowerId-int;

flowerName-String;

price-int;

rating-int;

type-String;

The above attributes must be private,write getters and setters and **parametrized constructor** as well.

Create class MyClass with main method.

Implement one static method:findMinPriceByType in MyClass class.

findMinPriceByType Method:

- ❖ This method will take **two input parameters:Array of Flower objects and String parameter**.This method will return the **minimum priced flower object** from **array of Flower objects** for the **given type(String parameter)**,only if the **rating is greater than**
- ❖ **3.** If no Flowers with the above conditions are present in the array of the flower objects,then the method will return null.

For this method-The **main method should print the flowerId from the returned object** if the returned value is not null.

else it should print "There is no flower with given type".

input1:

123

Yellow trout lilly

3000

5

ephemerals

345

snowdrop

2500

4

ephemerals

321

Yarrow

1000

4

perennials

213

red trillium

2250

4

ephemerals

ephemerals

output:

213

input2:

123

Yellow trout lilly

3000

5

ephemerals

345

snowdrop

2500

4

ephemerals

321

Yarrow

1000

4

perennials

213

red trillium

2250

4

ephemerals

Biennial

output2:

There is no flower with given type.

QUESTION 19

Create a class Engine with the below attributes:

engineId – int

engineName – String

engine Type – String

enginePrice – double

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class MyClass with main method.

Implement two static methods – findAvgEnginePriceByType and searchEngineByName in MyClass class.

findAvgEnginePriceByType method: This method will take **two input parameters Engine objects and String parameter**.

The method will **return the average price of Engines from array of Engine objects for the given engine type**

(String parameter passed). If no Engine with the given type is present in the array of Engine objects, then the method

should return 0.

searchEngineByName method:

- ❖ This method will take a String parameter and an array of Engine objects as the other parameter.
- ❖ The method will return **Engine object array in an ascending order of their engine ids**, from the array of Engine objects whose
- ❖ name attribute **matches with the given engine name** (String parameter passed). If no Engine with the given name is present in
- ❖ the array of Engine objects, then the method should return null.

Note: No two Engine object would have the same engine id.

All searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For findAvgEnginePriceByType method: The **main method** should **print the average enginePrice of Engines** as it is, if the returned

value is greater than 0, or it should print “There are no engine with given type”.

For searchEngineByName method: The main method should print the engineId from the returned Engine object array if the returned

value is not null. If the returned value is null, then it should print "There are no engine with the given name".

Before calling these static methods in main, use Scanner object to read the values of four Engine objects referring attributes

in the above mentioned attribute sequence. next, read the value of two String parameters for capturing engine type and engine

name respectively.

Input :

2
1001
Maruti
Diesel
20000
1002
Kia
Petro
17000
1003
Hyundai
Diesel
24000
1000
4
Maruti
Petrol
27500
Petrol → String engine type
Maruti → String engine name

Output :

27500

1000

1001

QUESTION 19

Create the **class Flowers with** below attribute:

flowerId-int;

flowerName-String;

price-int;

rating-int;

type-String;

The above attributes must be private,write getters and setters and parametrized constructor as well.

Create class MyClass with main method.

Implement one static method:findMinPriceByType in MyClass class.

findMinPriceByType Method:

- ❖ This method will take **two input parameters:Array of Flower objects** and **String parameter**.This method will return the **minimum**
- ❖ **priced flower object from array of Flower objects** for the given type(String parameter),only if the **rating is greater than 3**.
- ❖ If no Flowers with the above conditions are present in the array of the flower objects,then the method will return null.

For this method-The main method should print the flowerId from the returned object if the returned value is not null.

else it should print "There is no flower with given type".

input1:

123

Yellow trout lilly

3000

5

ephemerals

345

snowdrop

2500

4

ephemerals

321

Yarrow

1000

4

perennials

213

red trillium

2250

4

ephemerals

ephemerals

output:

213

input2:

123

Yellow trout lilly

3000

5

ephemerals

345

snowdrop

2500

4

ephemerals

321

Yarrow

1000

4

perennials

213

red trillium

2250

4

ephemerals

Biennial

output2:

There is no flower with given type.

QUESTION 21

Create a class Fruits with below attributes as private

- int fruitId
- String fruitName
- int price
- int rating

Write getters, setters and parameterized constructor in the abovementioned attribute sequence as required.

Create class Solution with main method

Implement One static method-

- **findMaximumPriceByRating**

findMaximumPriceByRating in the solution class.

This method will take **array of Fruit objects**, **int paramater** and returns the **maximum priced fruit object** from array of Fruit objects whichever has **rating greater than the rating** (rating parameter passed). This method should be called from main method.

Write code to perform following tasks:

1. Take necessary input variable and call findMaximumPriceByRating.

For this method- The main method should print the fruitId from returnedobject. As if the returned value is not null, or it should print "No such Fruit".

Note:- Before calling this static method, use Scanner object to read the values of four Fruits objects. No two Fruits will have same price. All these searches should be case insensitive.

Consider below sample input and output to test your code.

Input

555

Apple

200

5

777

Orange

150

4

333

Banana

100

4

888

Avocado

250

4

3

Output

888

Question 22

Create **class Associate** with below attributes:

id - int

name - String

technology - String

experienceInYears - int

Create class Solution and implement static method "associatesForGivenTechnology" in the Solution class.

This method will take **array of Associate objects** and a **searchTechnology String as parameters**.

And will **return another array of Associate objects** where the searchTechnology String matches with the original array of

Associate object's technology attribute (case insensitive search) and **experienceInYears attribute should be multiples of 5**.

Write necessary getters and setters.

Before calling "associatesForGivenTechnology" method in the main method, read values for five associate objects referring

the attributes in above sequence along with a String searchTechnology.

Then call the "associatesForGivenTechnology" method and write the logic to print the id's in the main method.

Input

101

Alex

Java

15

102

Albert

Unix

20

103

Alferd

Testing

13

104

Alfa

Java

15

105

Almas

Java

29

Java

Output

101

104

Question 23

Create a class Institution with below attributes:

institutionId - int

institutionName - String

noOfStudentsPlaced - int

noOfStudentsCleared- int

location - String

grade - String

Write getters, setters for the above attributes.

Create constructor which takes parameter in the above sequence except grade.

Create class Solution with main method.

Implement two static methods - FindNumClearancedByLoc and UpdateInstitutionGrade in Solution class.

FindNumClearancedByLoc method:

- This method will **take two input parameters - array of Institution objects and string parameter location.**
- The method will return the **sum of the noOfStudentsCleared attribute** from institution objects for the location passed as
- parameter. If no institution with the given location is present in the array of institution objects, then the method
- should return 0.

UpdateInstitutionGrade method:

This method will take a String parameter institutionName, along with the array of Institution objects.

The method will **return the institution object**, if the **input String parameter matches** with the institutionName attribute of the institution object.

Before returning the object, the **grade should be arrived based on the rating calculation mentioned below.**

This grade value should be assigned to the object. If any of the above conditions are not met, then the method should return null.

*** The grade attribute should be calculated as follows:**

$\text{Rating} = (\text{noOfStudentsPlaced} * 100) / \text{noOfStudentsCleared}$

If the rating ≥ 80 , then grade should be 'A'.

Else, then grade should be 'B'

****Note:****

No institution object would have the same value for institutionName attribute.

All institution object would have the noOfStudentsPlaced value lesser than noOfStudentsCleared value.

All the searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For FindNumClearancedByLoc method - The main method should print the noOfClearance as it is, if the returned value is greater than 0, or it should print "There are no cleared students in this particular location".

For UpdateInstitutionGrade method - The main method should **print the institutionName and grade of the returned Institution**

object. The institutionName and grade should be concatenated with :: while printing. eg:- TCS::A, where TCS is the

institution name and A is the grade.

If the returned value is null then it should print "No Institute is available with the specified name".

Before calling these static methods in main, use Scanner object to read the values of four Phone objects referring attributes

in the above mentioned attribute sequence (except grade attribute). Next, read the value for location and institutionName.

Input

111

Amrita

5000

10000

Chennai

222

Karunya

16000

20000

Coimbatore

333

AppleTech

10000

12000

Chennai

444

Aruna

6000

10000

Vellore

Chennai

Karunya

Output

22000

Karunya::A

Starter Code

=====

```
import java.util.Scanner
```

```
public class Solution
```

```
{
```

```
public static void main(String[] args)
{
    //code to read values
    //code to call required method
    //code to display the result
}
```

```
public static int findNumClearancedByLoc(Institution[] instArray, String location)
{
    //method logic
}
```

```
public static Institution updateInstitutionGrade(Institution[] instArray, String instName)
{
    //method logic
}
}

class Institution
{
    //code to build Institution class
}
```

Question 24

Create class Inventory with below attributes:

inventoryId - String

maximumQuantity - int

currentQuantity - int

threshold - int

Create class Solution and implement **static method "Replenish"** in the Solution class.

This method will take array of **Inventory objects** and a **limit int** as parameters.

And will **return another array of Inventory objects** where the **limit int** is **greater than or equal to the original array of**

Inventory object's threshold attribute.

Write necessary getters and setters.

Before calling "Replenish" method in the main method, read values for four Inventory objects referring the attributes in above

sequence along with a int limit. Then call the "Replenish" method and write logic in main method to print "Critical Filling",

if the threshold attribute is greater than 75. Else if the threshold attribute is between 50 and 75 then print

"Moderate Filling". Else print "Non-Critical Filling"

Input

1

100

50

50

2

200

60

40

3

150

35

45

4

80

45

40

45

Output

2 Non-Critical Filling

3 Non-Critical Filling

4 Non-Critical Filling

Question 25

Create a class Cinema with the below attributes:

movieId - int

director - String

rating - int

budget - int

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create class Solution with the main method.

Implement two static methods - findAvgBudgetByDirector and getMovieByRatingBudget in Solution class.

findAvgBudgetByDirector method:

This method will take two **input parameters** - **array of Movie objects** and **string parameter director**.

The method will **return the average of the budget attribute** from Movie objects directed by the **director passed as parameter**.

If no movie with the given director is present in the array of movie objects, then the method should return 0.

getMovieByRatingBudget method:

This method will **take two int parameters rating and budget**, along with the **array of movie objects**.

The method will **return the movie object**, if the **input parameters rating and budget**, matches with the **rating and budget** attribute of the **movie object respectively**.

Also check if rating is a factor of budget (eg: 3 is a factor of 12 because 3 divides 12 without leaving a remainder).

If any of the conditions are not met, then the method should return null.

Note : No two movie object would have the same value for rating and budget attributes.

All the searches should be case insensitive.

The budget mentioned are in crores and in INR.

These above mentioned static methods should be called from the main method.

For findAvgBudgetByDirector method - **The main method should print the average budget as it is if the returned value is**

greater than 0, or it should print "Sorry - The given director has not yet directed any movie".

For getMovieByRatingBudget method - **The main method should print the movieId** of the returned movie object. If the returned

value is null then it should print "Sorry - No movie is available with the specified rating and budget requirement".

Before calling these static methods in main, use Scanner object to read the values of four Movie objects referring attributes

in the above mentioned attribute sequence. Next, read the value for director, rating and budget.

Input

1101

GVM

4

100

1201

Shankar

5

500

1301

Shankar

3

50

1401

GVM

5

300

GVM

5

300

Output

200

1401

QUESTION 26

Create class Player2 with below attributes:

id - int

name - String

iccRank - int

matchesPlayed - int

runsScored - int

Create class Solution and implement static method "findAverageOfRuns" in the Solution class.

- This method will take **array of Player objects and a target int** as parameters.
- And will **return another double array** where **the target int is lesser than or equal to the original array of Player object's matchesPlayed attribute** and contains **the average run scored by each player satisfying above condition**.

Write necessary getters and setters.

Before calling "findAverageOfRuns" method in the main method, read values for four Player objects referring the attributes in above sequence along with a int target.

Then call the "findAverageOfRuns" method and write logic in main method **to print "Grade A", if the calculated averageRun value is 80 to 100**. Else if **the averageRun value is between 50 and 79** then **print "Grade B"**. Else print **"Grade C"**.

Input

100

Sachin

5

150

13000

101

Sehwag

4

120

10000

103

Dhoni

7

110

7000

104

Kohli

15

57

4400

100

Output

Grade A

Grade A

Grade B

QUESTION 27

Create class Sim2 with below attributes:

id - int

company – String

balance - int

ratePerSecond - double

circle - String

Create class Solution and implement **static method "matchAndSort" in the Solution** class.

- This method will take **array of Sim objects, search_circle String and search_rate double** as parameters.
- And will return **another Sim array where the search_circle matches with the circle parameter of the original Sim array and**
- the **search_rate double is greater than the original array of Sim object's ratePerSecond attribute which is also sorted by means of balance attribute in descending order.**

Write necessary getters and setters.

Before calling "matchAndSort" method in the main method, read values for four Sim objects referring the attributes in above sequence along with a String search_circle and a double search_rate. Then call the "matchAndSort" method and write logic in

main method to print the id's of the result obtained.

Input

1

jio

430

1.32

mumbai

2

idea

320

2.26

mumbai

3

airtel

500

2.54

mumbai

4

vodafone

640

3.21

mumbai

mumbai

3.4

Output

4

3

1

2

QUESTION 28

Create class Team with below attributes:

tId - int

tName - String

tCountry - String

tRun - int

The above attributes should be private. Write Getter and Setter and parametrized constructor as required.

Create class Solution with main method.

Implement One static method-

- **findPlayer**

findPlayer in the solution class.

This method will take array of **Team objects**, **int paramater** run and **String country** and returns **the Team object** where taken **country** is matched with the **country of team object** and **team run is greater than the taken run**.

Write code to perform following tasks:

1. In main mehod, take 1st input as integer parameter which shows the number of team object.
2. Take necessary input variable and call findPlayer.
3. **All searches should be case insensitive.**

For this method- **The main method should print the team object**. As if the returned value is not null,

or it should print "No team is found from the given country and run".

Input

3

1

Arijit

India

43

2

Doremon

China

40

3

Donal

England

32

35

China

Output

2

Doremon

China

40

QUESTION 29

Find **total interest of an amount.**

Create a class Account with following attributes :

id - int

balance - double

interestRate - double

Take **another integer as number of year.**

After that **calculate total interest using formula:**

Percentage = $\text{interestRate} * 100 / \text{time}$

Total Interest = Percentage + interestRate

QUESTION 29

Create **class Document** with below attributes

id - int

title - String

folderName - String

pages - int

Write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement static method - **docsWithOddPages** in Solution class.

This method will take **array of Document objects** and **return another array with Document objects which has odd number of pages**.

This method should be called from main method and **display values of returned objects as shared** in the sample **(in ascending order of id attribute)**.

Before calling this method, use Scanner object to read values for four Document objects referring attributes in the above sequence.

Next call the method and display the result.

Consider below sample input and output:

Input:

1

resume

personal

50

2

question1

exams

55

3

question2

exams

45

4

India

misc

40

Output (each line has values separated by single space):

2 question1 exams 55

3 question2 exams 45

Note on using Scanner object:

Sometimes scanner does not read the new line character while invoking methods like `nextInt()`, `nextDouble()` etc.

Usually, this is not an issue, but this may be visible while calling `nextLine()` immediately after those methods.

Consider below input values:

22

hello

Referring below code:

```
Scanner sc = new Scanner(System.in);
```

```
int x = sc.nextInt();
```

`String str = sc.nextLine();` -> here we expect str to have value hello. Instead it may be `""`.

If above issue is observed, then it is suggested to add one more explicit call to `nextLine()` after reading numeric value.

QUESTION 30

Create class Document with below attributes

id - int

title - String

folderName - String

pages - int

Write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement static method - docsWithOddPages in Solution class.

This method will take array of Document objects and return another array with Document objects which has odd number of pages.

This method should be called from main method and display values of returned objects as shared in the sample (in ascending order of id attribute).

Before calling this method, use Scanner object to read values for four Document objects referring attributes in the above

sequence.

Next call the method and display the result.

Consider below sample input and output:

Input:

1

resume

personal

50

2

question1

exams

55

3

question2

exams

45

4

India

misc

40

Output (each line has values separated by single space):

2 question1 exams 55

3 question2 exams 45

Note on using Scanner object:

Sometimes scanner does not read the new line character while invoking methods like `nextInt()`, `nextDouble()` etc.

Usually, this is not an issue, but this may be visible while calling `nextLine()` immediately after those methods.

Consider below input values:

22

hello

Referring below code:

```
Scanner sc = new Scanner(System.in);
```

```
int x = sc.nextInt();
```

`String str = sc.nextLine();` -> here we expect str to have value hello. Instead it may be `""`.

If above issue is observed, then it is suggested to add one more explicit call to `nextLine()` after reading numeric value.

QUESTION 31

Create a class Music with below attributes:

playListNo - int

type -String

count - int

duration(minutes) -double.

The above attributes should be private. Write Getter and Setter and parametrized constructor as required.

Create class Solution with main method.

Implement two static methods: findAvgOfCount and sortByDuration in Solution Class.

findAvgOfCount Method:

- This method will take an **array of Music objects** and **an int value** as input parameters.
- The method will **find out Average (as int) of count for those objects whose count is more than the given Count(int parameter passed)**. This **method will return average** if found. If there is no Type that matches then the method should return 0.
- for (int i = 0; i < musicArray.length; i++)
- { Music music = musicArray[i];
- if (music.getCount() > minCount)
- { sum += music.getCount(); // Add to sum if count is greater than minCount count++; // Increment valid music count } } // Calculate average if count of valid music is greater than 0
- if (count > 0)
- { return sum / count; // Return the average }

- else { return 0; // Return 0 if no valid music found } }

for this method- main method should print average ,if the returned value is not 0.If the returned value is 0 then main method

should print "No playlist found".

sortTypeByDuration method:

This method will take an **array of Music objects** and an **int value** as input parameters.This method should return an array of Music objects in an **ascending order of their duration** which are **more than the given duration** (int parameter passed) value.

If there are no such objects ,then the method should return null.

for this method-The main method should print the type from the returned Music object array if the returned value is not null.

If the returned value is null then the main method should print "No playlist found with mentioned attribute".

Consider the below input and output:

input1:

111

WorkOut

10

15.2

321

Dance Party

20

55.500

721

Childhood Jams

6

50.60

521

Chill

30

78.89

15

20

output1:

25

Childhood Jams

Dance Party

Chill

input2:

111

Oldies but Goodies

17

55

321

Guilty Pleasures

27

27

721

night drive

21

345

521

Rainy day

34

21

50

5000

output 2:

No playlist found.

No playlist found with mentioned attribute.

QUESTION 32

Create a **class TravelAgencies** with below attributes:

regNo – int

agencyName – String

packageType – String

price – int

flightFacility – boolean

Write getters, setters for the above attributes . Create constructor which takes parameter in the above sequence.

Create class Solution with main method. Implement two static methods – **findAgencyWithHighestPackagePrice** and **agencyDetailsforGivenIdAndType** in Solution class.

findAgencyWithHighestPackagePrice method:

This method will take **array of TravelAgencies objects** as an **input parameter** and return the **highest package price** from the given array of objects.

agencyDetailsForGivenIdAndType method:

This method will take **three input parameters** -array of TravelAgencies objects, int parameter **regNo** and String parameter **packageType**. The method will **return the TravelAgencies object** based on below conditions.

- **FlightFacility** should be available.
- The **input parameters(regNo and packageType)** should **matched** with the **regNo and packageType** of TravelAgencies object.
- If any of the above conditions are not met, then the method should return null. Note : **Same Travel agency can have more than one package type**. **Travel agency and package type combination is unique**. All the searches should

be case insensitive.

The above mentioned static methods should be called from the main method.

For **findAgencyWithHighestPackagePrice** method – **The main method** should **print the highestPackagePrice** as it is.

For **agencyDetailsForGivenIdAndType** method -The main method should **print the AgencyName** and **price** of the returned

object. The AgencyName and price should be concatenated with : while printing.

Input

123

A2Z Agency

Platinum

50000

true

345

SSS Agency

Gold

30000

false

987

Cox and Kings

Diamond

40000

true

888

Global Tours

Silver

20000

false

987

Diamond

Output

50000

Cox and Kings:40000

QUESTION 33

Create a class newspaper with below attributes.

regNo-int

name-String

publicationYear-int

price-int

Write parametrised constructor as required.

Create class Solution with main method.

Implement 2 static methods **findTotalPriceByPublicationYear** and

searchNewspaperByName in solution class.

findTotalPriceByPublicationYear method:

- This method will take **array of the Newspaper objects** and **int parameter** typ.
- This method will **return the total price of newspapers** from array of Newspaper objects,
- If the publication **year attribute matches with the int parameter passed** .If no Newspaper with the given publication year is present in the array ,then the method should return 0.
- The main method **should print total price if the returned value is greater than 0**.Else it should print "No Newspaper found with the mentioned attribute".

searchNewspaperByName method :

This method will take **the array of the Newspaper objects** and **the String parameter type**.This method will **return the Newspaper object** from the Array of Newspaper objects,If the name **attribute matches with the string parameter passed**.

If no Newspaper with the given name is present in the array ,then the

method should return null.

The main method should print the regNo,name,year of publication,and price

if the returned Newspaper object is not null.

else it should print "No Newspaper found with the given name."

The above mentioned static methods should be called from main method.

```
*****  
*****
```

input1:

134

Evening Mail

2018

79

178

Morning Mail

2017

65

176

Daily News

2017

98

219

Times of India

1988

15

2017

Times of India

output1:

163

regNo-219

name-Times of India

publicationYear-1988

price-15

input2:

201

Afternoon Mail

2019

79

202

Morning Mail

2014

65

156

Daily News

2004

98

278

Times of India

1967

15

2017

Times of America

output2:

No Newspaper found with the mentioned attribute

No Newspaper found with the given name

5

Create class Player3 with below attributes:

id - int

name - String

Country - String

matchesPlayed - int

runsScored - int

the above attribute should be private ,write getter and setter and parametrized constructor as well.

- Create **class Solution** and implement **static method "findPlayerName"** in the Solution class.
 - This method **will take array of Player objects** and a **String as country parameters** and **int as run parameter**.
 - And will **return another array of Player object** where the **String is matched with country** and **run scored of that Player object is greater than the input int parameter**. **All the searches should be case insensitive**.
-
- Before calling "findPlayerName" method in the main method, read values for Player objects **referring the attributes in above sequence along with a int target**.
 - Then call the "findPlayerName" method and write logic in main method to print Player object. If any of the input doesn't match
 - with the following attributes then print "No player found".

Note:

1. First take input as the number of player object, then the player details

2. **Show the output as the dictionary order of the players name**

3. **First print player id, and then use ' : ' sign and after that player name**

3 : Arijit

1 : Minaz

5 : Priyanka

Input : 1

5

1

Arnab

India

4

94

2

Virat

UK

5

68

3

Arijit

India

82

87

4

Raj

India

21

78

5

Priyanka

India

12

86

India

80

Output : 1

3:Arijit

1:Arnab

5:Priyanka

Input : 2

5

1

Arnab

India

4

94

2

Virat

UK

5

68

3

Arijit

India

82

87

4

Raj

India

21

78

5

Priyanka

India

12

86

UK

80

Output : 2

No player found

QUESTION 35

Create a class Car with the below attributes:

carId – int

carName – String

fuelType – String

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class RemoveCar with main method.

Implement two static methods – **RemoveAndRearrange in RemoveCar class.**

RemoveAndRearrange method:

This method will take **int input parameter**. The method **will return the new array of Car object and remove that car whose id is match with the input parameter**. **Remove the car details and return the array with continuous new ids.**

If no car with the given id is present in the array of Car objects, then the method should return null.

Note: No two Car object would have the same engine id.

The above mentioned static methods should be called from the main method.

For RemoveAndRearrange method: The main method should **print the id and car name as it is**, if the returned value is not null, or it should print “There are no car with given id”.

Before calling these static methods in main, use Scanner object to read the values of five objects referring attributes in the above mentioned attribute sequence. next, read the value of int parameters for capturing car id.

Input :

1001

Maruti

Diesel

1002

Kia

Petro

1003

Hyundai

Diesel

1004

Taxi

Petrol

1005

Truck

Diesel

1003

Output :

1001 : Maruti

1002 : Kia

1003 : Hyundai

1004 : Taxi

QUESTION 36

Create a class MobileDetails with the below attributes:

MobileId – int

Price - int

Brand – String

isFlagShip – boolean

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class MobileDetails with main method.

Implement two static methods – getTotalPrice() and getSecondMin() in MobileDetails class.

getTotalPrice method:

This method will take **two int input parameters which take object array and string brand**. The method will return **the totalprice of mobiles of that brand**.

If no mobile with the given brand is present in the array of MobileDetails objects, then the method should return 0.

getSecondMin method:

This method will **take object array which will return the second minimum price from the array and name of that mobile brand**.

The above mentioned static methods should be called from the main method.

For getTotalPrice method: The main method should print the total price as it is, if the returned value is not null, or it should print "There are no mobile with given brand".

Before calling these static methods in main, use Scanner object to read the values of five objects referring attributes

in the above mentioned attribute sequence.

Input :

101

25000

Samsung

True

102

19000

Redmi

False

103

28000

Nokia

True

104

25000

Realme

False

105

40000

Samsung

True

Samsung

Output :

65000

Samsung : 25000

Realme : 25000

QUESTION 37

create the class Song with below attributes.

songId - int

title - String

artist - String

duration-double

The above attributes should be private, getters, setters and parameterized constructor as required.

Create class MyClass with main method. **Implement two static methods - findSongDurationForArtist and getSongsInAscendingOrder in MyClass class.**

findSongDurationForArtist method:

This method will take **two input parameters of Song objects and String parameter.**

The method will return the **sum of song duration from array of Song object for the given artist (String parameter passed).**

If no Song with the given artist is present in the array of Song objects, then the method should return zero.

getSongsInAscendingOrder method:

This method will take input parameters array, of Song objects and String parameter. The method will return **Song objects array in an ascending order of their duration**, from **the array of Song objects whose artist attribute matches with the input String parameter**. If no Song with the given artist is present in the array of Song objects, then the method should return null.

Note:

1. No two Song object would have the same songid.
2. Combination of artist and duration should be unique.
3. All the searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For getfindSongDuration ForArtist method - in method should print the sum of duration of artist as it is, if the returned

value is greater than 0 or it should print "There are no songs with given artist".

For getSongsinAscendingOrder method - **The main method should print the song id and title from the** returned song object array

if the returned value is not null. If the returned value is null then it should print "There are no songs with given artist".

For Example,

112

ABC

where 112 is song Id. ABC is song title.

Before calling these static methods in main, use. Scanner object to read the values of five Song objects referring attributes

in the above mentioned attribute sequence. Next, read two String values for capturing artists.

Sample Input 1:

2150

In time

Justin Timberlake

4

250

Cry Me

Justin Timberlake

3

1200

Mirrors

Justin Timberlake

5

1300

That's the way it is

celion dion

5

500

Ashes

celion dion

3

celion dion

Justin Timberlake

Sample output1:

8.0

250

Cry Me

2150

In time

1200

Mirrors

Sample Input 2:

2150

Cry Me

Justin Timberlake

3

1000

Why Not

Enrique Iglesias

5

1200

Mirrors

Justin Timberlake

5

1300

That's the way it is

Celion Dion

5

500

Ashes

celion Dion

3

Bryan Adams

Michael Larkson

Output 2:

There are no songs with given artist

There are no songs with given artist

QUESTION 38

Create a class HeadSets with below attributes:

headsetName - String

brand-String

price - int

available- boolean

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement two static methods - findTotalPriceForGivenBrand and findAvailableHeadsetWithSecondMinPrice in Solution class.

findTotalPriceForGivenBrand method:

The method will return the **total price of Headsets from array of Headset objects for the given brand**(String parameter passed).

If no Headsets with the given brand is present in the array of Headsets objects, then the method should return 0.

findAvailableHeadsetWithSecondMinPrice method:

This method will **take array of Headset objects as an input parameter** and **returns the available Headset object** with the **second lowest/minimum price from the given array of objects**.

If no Headsets with the above condition is present in the array of Headsets objects, then the method should return null.

Note :

No two Headsets will have the same price.

All the searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For findTotalPriceForGivenBrand method - The main method should print the returned total price as it is if the returned value is greater than 0 or it should print "No Headsets available with the given brand".

Eg: 4500

where 4500 is the total Headsets price

For findAvailableHeadsetWithSecondMinPrice method - The main method should print the headsetsName and price from the

returned Headsets object if the returned value is not null.

If the returned value is null then it should print "No Headsets available".

Eg:Logitech Wired

1500

where Logitech Wired is the headsetName and 1500 is the price

Before calling these static methods in main, use Scanner object to read the values of four Headset objects referring attributes in the above mentioned attribute sequence.

Next, read the value of String parameter for capturing brand.

Consider below sample input and output:

Input1:

boAt BassHeads

boAt

1220

true

Over Ear Wired

boAt

549

true

In Ear with Mic

JBL

450

true

Buds 2 Neo

RealMe

500

true

boAt

Output:

1769

Buds 2 Neo

500

Input2:

boAt BassHeads

boAt

1220

false

Over Ear Wired

boAt

1549

false

In Ear with Mic

JBL

450

false

Buds 2 Neo

RealMe

500

false

boAt

Output:

2769

No Headsets available

Sample code snippet for reference:

Please use below code to build your solution.

import java.util.Scanner;

public class MyClass
{

public static void main(String[] args)
{

//code to read values

//code to call required method

//code to display the result

```

}

public static int findTotalPriceForGivenBrand (/* required parameters to be added */)
{
    Restricted for circulation outside TCS Xplore 9
    //method logic
}

public static HeadSets findAvailableHeadsetWithSecondMinPrice (/* required parameters to be
added */)
    //method logic
}

class HeadSets
{
    //code to build the class
}

```

Note on using Scanner object:

Sometimes scanner does not read the new line character while invoking methods like nextInt(), nextDouble() etc.

Usually, this is not an issue, but this may be visible while calling nextLine() immediately after those methods.

Consider below input values:

1001

Savings

Referring below code:

```
Scanner sc = new Scanner(System.in);
```

```
int x = sc.nextInt();
```

```
String str = sc.nextLine(); -> here we expect str to have value Savings. Instead it may be "".
```

If above issue is observed, then it is suggested to add one more explicit call to nextLine() after reading numeric value.

QUESTION 39

Create a class Vegetable with below attributes:

VegetableId - int

VegetableName - String

Price - int

Rating - int

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement two static methods - findMinimumPriceByRating in Solution class.

findMinimumPriceByRating method:

The method will return **minimum price object** from array of vegetable objects, whichever by **rating greater than given rating.**

If no vegetable with the above condition is present in the array of vegetable object, then the method return null.

Note :

No vegetable has the same price.

All the searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For findMinimumPriceByRating method - The main method should print the id of vegetable object or it should print "No such Vegetables".

Eg: 1002

where 1002 is the id whose price is lowest

Before calling these static methods in main, use Scanner object to read the values of four Vegetable objects referring attributes in the above mentioned attribute sequence.

Next, read the value of int parameter for rating.

Consider below sample input and output:

Input1

1001

Carrot

90

5

1002

Tomato

40

4

1003

Bectroot

80

4

1004

Onion

78

3

2

Output1

1002

=====

Input2

1001

Carrot

90

5

1002

Tomato

40

4

1003

Bectroot

80

4

1004

Onion

78

3

7

Output2

No such Vegetables

QUESTION 40

Create a **class Car** with the following attributes:

make - String

model - String

year - int

price - double

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class Solution with the main method.

Implement two static methods - findMostExpensiveCar and getCarByMakeAndModel in Solution class.

findMostExpensiveCar method:

This method will take an array of Car objects as input.

The method will **return the Car object with the highest price.**

If the array is empty, the method should return null.

getCarByMakeAndModel method:

This method will take **three input parameters - array of Car objects, make and model (both strings).**

The method will return the **Car object that matches the make and model attributes passed as parameters.**

If no car with the given make and model is present in the array of car objects, then the method should return null.

Note: The search should be case insensitive.

These above mentioned static methods should be called from the main method.

For findMostExpensiveCar method - The main method should print the make, model, year and price of the returned Car object.

If the returned value is null then it should print "Sorry - No car is available".

For getCarByMakeAndModel method - The main method should print the year and price of the returned Car object. If the returned

value is null then it should print "Sorry - No car is available".

Before calling these static methods in main, use Scanner object to read the values of four Car objects referring attributes in the above mentioned attribute sequence. Next, read the values for make and model.

Input:

Mercedes

S-Class

2020

1250000

BMW

5-Series

2019

900000

Audi

A4

2022

800000

Jaguar

F-PACE

2021

1100000

Audi

A4

Output:

Mercedes

S-Class

1250000.0

2020

2022

800000.0

QUESTION 41

Create a class **BankAccount** with the following attributes:

accountNumber - int

accountHolderName - String

balance - double

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class Solution with the main method.

Implement two static methods - withdraw and deposit in Solution class.

withdraw method:

- This method will **take two input parameters - array of BankAccount objects and the account number (int)** from which withdrawal should be done.
- The method will **check if the account number is present in the array of BankAccount objects.**
- If present, **the method will then check if the balance is sufficient to withdraw the requested amount (double).**
If yes, it will -
- **deduct the amount from the account balance and return the updated balance.**
- If the **account number is not present or the balance is not sufficient, the method should return -1.**

deposit method:

- This method will take **two input parameters** - array of BankAccount objects and the account number (int) in which deposit should be done.
- The method will **check if the account number is present in the array of BankAccount objects**.
- If present, **it will add the deposit amount (double) to the account balance and return the updated balance**.
- **If the account number is not present, the method should return -1.**

These above mentioned static methods should be called from the main method.

For withdraw method - The main method should print the updated balance if the returned value is greater than or equal to 0, or it should print "Sorry - Insufficient balance" if the returned value is -1, or it should print "Sorry - Account not found" if the returned value is -2.

For deposit method - The main method should print the updated balance if the returned value is greater than or equal to 0, or it should print "Sorry - Account not found" if the returned value is -1.

Before calling these static methods in main, **a parameterized constructor in the above mentioned attribute sequence as required.**

Input

1001
Alice
5000.0
1002
Bob
10000.0
1003
Charlie
15000.0
1002
5000.0
1001
10000.0

Output

5000.0

15000.0

Question 42

Create a class **Person** with the following attributes:

firstName - String

lastName - String

age - int

gender - char ('M' for male and 'F' for female)

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class **Student** that extends the **Person** class and adds the following attributes:

rollNo - String

course - String

semester - int

GPA - double

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class **Faculty** that extends the **Person** class and adds the following attributes:

employeeId - String

department - String

designation - String

salary - double

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class Solution with the main method.

Implement two static methods - findHighestGPASStudent and findHighestPaidFaculty in Solution class.

findHighestGPASStudent method:

- This method will take an array of Student objects as input.
- The method will **return the Student object with the highest GPA.**
- If the **array is empty, the method should return null.**

findHighestPaidFaculty method:

- This method will take **an array of Faculty objects as input.**
- The method will return **the Faculty object with the highest salary.**
- If the **array is empty, the method should return null.**

These above mentioned static methods should be called from the main method.

For findHighestGPASStudent method - The main method should print the roll number, course and GPA of the returned Student object.

If the returned value is null then it should print "Sorry - No student is available".

For findHighestPaidFaculty method - The main method should print the employee ID, department and salary of the returned

Faculty object.

If the returned value is null then it should print "Sorry - No faculty is available".

Before calling these static methods in main, use Scanner object to read the values of two Student objects and two Faculty objects referring attributes in the above mentioned attribute sequence.

Example Input:

First1

Last1

20

M

Roll1

Course1

2

3.8

First2

Last2

21

F

Roll2

Course2

3

3.9

Faculty1

Last3

35

M

ID1

Department1

Associate Professor

90000.0

Faculty2

Last4

40

F

ID2

Department2

Professor

100000.0

Output:

Roll2

Course2

3.9

ID2

Department2

100000.0

Question 43

Create a **class Student** with the following attributes:

name - String

rollNo - String

age - int

gender - char ('M' for male and 'F' for female)

course - String

semester - int

GPA - double

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class Solution with the main method.

Implement two static methods - **calculateAverageGPA** and **getStudentsByCourse** in Solution class.

calculateAverageGPA method:

- This method will **take an array of Student objects as input.**
- The method will **return the average GPA of all the students in the array.**
- **If the array is empty, the method should return 0.**

getStudentsByCourse method:

- This method will take **two input parameters - array of Student objects and a course (string).**
- The method will return **an array of Student objects that are enrolled in the specified course.**
- If no student with the given course is present in the array of Student objects, then the method should return an empty array.

These above mentioned static methods should be called from the main method.

For calculateAverageGPA method - The main method should **print the average GPA returned** by the method.

If the returned value is 0, then it should print "Sorry - No students are available".

For getStudentsByCourse method - The main method should print the **name, roll number and GPA of the returned Student objects**.

If the returned value is an empty array, then it should print "Sorry - No students are available for the given course".

Before calling these static methods in main, use Scanner object to read the values of four Student objects referring

attributes in the above mentioned attribute sequence. Next, read the value for a course.

Example Input:

John

1234

20

M

CSE

2

3.5

Alice

5678

19

F

EEE

1

3.7

Bob

9101

21

M

CSE

4

3.9

Sarah

2345

20

F

ME

3

3.2

CSE

Example Output:

3.6

John

1234

3.5

Bob

9101

3.9

QUESTION 44

Create a class Rectangle with the following attributes:

length - double

breadth - double

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class Solution with the main method.

Implement two static methods - calculateArea and calculatePerimeter in Solution class.

calculateArea method:

This method will take a **Rectangle object as input**.

The method will calculate the **area of the rectangle and return it as a double**.

calculatePerimeter method:

This method will take a **Rectangle object as input**.

The method will **calculate the perimeter of the rectangle and return it as a double**.

These above mentioned static methods should be called from the main method.

For calculateArea method - The main method should print the area returned by the method.

For calculatePerimeter method - The main method should print the perimeter returned by the method.

Before calling these static methods in main, use Scanner object to read the values of two Rectangle objects referring

attributes in the above mentioned attribute sequence.

Example Input:

5.5

3.2

7.0

4.5

Example Output:

17.6

17.4

31.5

23.0

QUESTION 45

Create a **class Employee** with the following attributes:

name (string)

age (int)

salary (double)

Write getters, setters, and a parameterized constructor in the above-mentioned attribute sequence as required.

Create a class Solution with the main method.

Implement two static methods - **calculateYearlySalary** and **calculateTax** in the Solution class.

✓ **calculateYearlySalary method:**

- This method will take an **Employee object as input**.
- The method will **calculate the yearly salary of the employee** (assuming that the employee works for 12 months in a year) and **return it as a double**.

✓ **calculateTax method:**

This method will take an Employee object as input.

The method will **calculate the tax to be paid by the employee** based on the following rules:

- If the yearly salary is less than or equal to 50000, the tax is 10% of the yearly salary.
- If the yearly salary is greater than 50000 but less than or equal to 100000, the tax is 20% of the amount over 50000 plus 10% of the first 50000.
- If the yearly salary is greater than 100000, the tax is 30% of the amount over 100000 plus 20% of the amount between 50000 and 100000 plus 10% of the first 50000.

- The method will return the tax amount as a double.

These above-mentioned static methods should be called from the main method.

For calculateYearlySalary method - The main method should print the yearly salary returned by the method.

For calculateTax method - The main method should print the tax amount returned by the method.

Before calling these static methods in main, use a Scanner **object to read the values of an Employee object's attributes.**

Example Input:

John

30

55000.0

Example Output:

Yearly salary of John: 660000.0

Tax to be paid by John: 183000.0

Question 21

Create a class Engine with the below attributes:

engineId – int

engineName – String

engine Type – String

enginePrice – double

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class MyClass with main method.

Implement two static methods – findAvgEnginePriceByType and searchEngineByName in MyClass class.

findAvgEnginePriceByType method: This method will take two input parameters Engine objects and String parameter.

The method will return the average price of Engines from array of Engine objects for the given engine type (String parameter passed). If no Engine with the given type is present in the array of Engine objects, then the method should return 0.

searchEngineByName method: This method will take a String parameter and an array of Engine objects as the other parameter.

The method will return Engine object array in an ascending order of their engine ids, from the array of Engine objects whose

name attribute matches with the given engine name (String parameter passed). If no Engine with the given name is present in

the array of Engine objects, then the method should return null.

Note: No two Engine object would have the same engine id.

All searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For findAvgEnginePriceByType method: The main method should print the average enginePrice of Engines as it is, if the returned

value is greater than 0, or it should print "There are no engine with given type".

For searchEngineByName method: The main method should print the engineId from the returned Engine object array if the returned

value is not null. If the returned value is null, then it should print "There are no engine with the given name".

Before calling these static methods in main, use Scanner object to read the values of four Engine objects referring attributes in the above mentioned attribute sequence. next, read the value of two String parameters for capturing engine type and engine name respectively.

Input :

1001

Maruti

Diesel

20000

1002

Kia

Petro

17000

1003

Hyundai

Diesel

24000

1000

Maruti

Petrol

27500

Petrol

Maruti

Output :

27500

1000

1001

Question 46

Define two functions as given below.

1. **validateScore**: This function will validate a given number whether it is less than or equal to 100 and greater than 0.

If these conditions are matched, function will return True, otherwise return False

2. **findValidScores**: This function will take a list of scores (numbers) as input, and return a list of scores from the input list after validation . Use the method validateScore to validate scores in the input list. If the input list does not contain any valid score, function should return None.

Define the main section for the following:

1. Create a list of positive integers by taking user inputs

To create the list, do the following:

a. First read the number of elements (integer) you want to store in the list as input from user

b. Read a number as input and add to the list. Repeat this step for the number of elements to be stored in the list (considered in the first line of input)

2. Call the method findValidScores by passing the list created in point #1 as argument.

3. Print a message "Valid Scores =" (excluding the quotes) followed by the list returned by the method findValidScores in the same line. If the method returns None, print the message "No valid score found." (excluding the quotes).

Sample Input description:

The first input is the number of elements to be added to the list.

The next set of inputs are the numbers to be added to the list one after another and is repeated for the number of elements

given in the first line of input.

Check the sample input and output below.

Sample Input 1:

For example, if you want to create a list of 5 positive numbers with values 10,20,101,90 and 0 then the input should be read/given as follows:

5

10

20

101

90

0

And the output for the above set of input will be:

Valid Scores = [10, 20, 90]

Sample Input 2:

8

-100

101

0

.80

101

120

-75

0.78

Output:

No valid score found.

Question 49

Create a **class Bill** with below attributes:

billNo- int

name - String

typeOfConnection - String

billAmount - double

status – boolean

where billNo is the bill number, name is the name of the customer, typeOfConnection is the type of the connection (prepaid, postpaid), billAmount is the bill amount and **status is whether the bill is paid or not (if paid then value is TRUE**
else value is FALSE).

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement two static methods - findBillWithMaxBillAmountBasedOnStatus and getCountWithTypeOfConnection in Solution class.

findBillWithMaxBillAmountBasedOnStatus method:

This method will take an **array of Bill objects** and a **boolean parameter as parameters.**

The method will return bill **object array in ascending order of their bill number** from the array of Bill objects **whose bill amount is maximum** in the array with the **status attribute** that matches **with the input parameter.**

If **no Bill** with **the given status** is present in the array of Bill objects, then the method should return null.

getCountWithTypeOfConnection method:

This method will take two input parameters - **array of Bill objects and string parameter (for type of connection).**

The method will return the **count of bills from array of bill objects** for the **given type of connection**.

If no bill with the given type of connection is present in the array of bill objects, then the method should return 0.

Note :

Two bill object can have the same bill amount.

All the searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For findBillWithMaxBillAmountBasedOnStatus method - The main method should print the **billNo followed by # and name from the returned Bill object** array if the returned value is not null.

If the returned value is null then it should print "There are no bill with the given status".

For getCountWithTypeOfConnection method - The main method should print the count of bills as it is, if the returned value is

greater than 0, otherwise it should print "There are no bills with given type of connection".

Before calling these static methods in main, use Scanner to read the number of object and objects to read the values of Bill

objects referring attributes in the above mentioned attribute sequence.

Next, read the value for status and typeOfConnection.

Consider below sample input and output:

Input:

4

111

Aman Mittal

Prepaid

914.25

true

222

Rekha Kumar

Prepaid

1425.75

false

333

Samyra Gupta

Prepaid

1305.00

true

444

Mohit Saxena

Postpaid

1300.50

false

false

Prepaid

Output:

222#Rekha Kumar

3

Sample code snippet for reference:

Please use below code to build your solution.

```
import java.util.Scanner;
```

```
public class Solution
```

```
{
```

```
    public static void main(String[] args)
```

```
    {
```

```
        //code to read values
```

```
        //code to call required method
```

```
        //code to display the result
```

```
}
```

```
public static Bill[] findBillWithMaxBillAmountBasedOnStatus( /* required  
parameters to be added */ ) {
```

```
//method logic
```

```
}
```

```
public static int getCountWithTypeOfConnection( /* required parameters to be  
added */ ) {
```

```
//method logic
```

```
}
```

```
}
```

```
class Bill
```

```
{
```

```
//code to build the class
```

```
}
```

Note on using Scanner object:

Sometimes scanner does not read the new line character while invoking methods like `nextInt()`, `nextDouble()` etc.

Usually, this is not an issue, but this may be visible while calling `nextLine()` immediately after those methods.

Consider below input values:

1234

Merin Bakers

Referring below code:

```
Scanner sc = new Scanner(System.in);
```

```
int x = sc.nextInt();
```

String str = sc.nextLine(); -> here we expect str to have value Bakers name.
Instead it may be "".

If above issue is observed, then it is suggested to add one more explicit call to nextLine() after reading numeric value.

Question 52

Create a **class Employee** with the below attributes:

EmpId - int

EmpName - String

Dept - String

Rating - int

Salary - int

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create class Solution with the main method.

Implement **two static methods** - **findDept** and **findSecondHighestSalary** in Solution class.



findDept method:

This method will take two input parameters - **array of Employee objects** and **int parameter**. The method

will **return the dept array attribute from employee objects** whose **salary is matched with int parameter and rating greater than** equal to 3. If no data found of beach objects, then the method should return null.

findSecondHighestSalary method:

This method will take **two input parameters** - **array of Employee objects** and **string parameter**. The method will return the **second highest salary** from employee objects whose **dept is matched with string parameter**.

If no data found of beach objects, then the method should return 0.

Note: No two employees have same rating. All the searches should be case insensitive.

These above mentioned static methods should be called from the main method.

For findDept method - The main method should print the dept from the Employee array. If return null, then you should print

"No Department found".

For findSecondHighestSalary method - The **main method should print** the **2nd highest salary** from the Employee array with the following condition. If return 0, then you should print "No data found".

Input ->

101

Arijit

Computer

4

35000

102

Rakesh

Electronics

2

18000

103

Mahima

Mechanical

3

35000

104

Saniya

Mechanical

5

68000

105

Rajesh

Computer

1

30000

35000

Mechanical

Output ->

Computer, Mechanical

35000

Question 50

Create a class called BankAccount with the following attributes:

accountNumber (String)

accountHolderName (String)

balance (double)

Write getters, setters, and a parameterized constructor for the class.

Create a static method transferFunds in a separate class called BankUtils.
The method should take four parameters:

fromAccount (BankAccount)

toAccount (BankAccount)

amount (double)

transactionCode (String)

The method should transfer the specified amount from the fromAccount to the toAccount,

and return a Transaction object containing the transactionCode, the amount

transferred, and the timestamp of the transaction. If the fromAccount has insufficient

balance, the method should throw an InsufficientBalanceException with an appropriate error message.

Create a class called Transaction with the following attributes:

transactionCode (String)

amount (double)

timestamp (String)

Write getters and a parameterized constructor for the class.

Create an InsufficientBalanceException class that extends the Exception class.

The exception should have a constructor that takes a message as a parameter.

In the main method, create two BankAccount objects, transfer some funds between

them using the transferFunds method, and print the transaction details to the console. Handle any exceptions.

Sample Input:

Account 1:

Account Number: 12345

Account Holder Name: John

Balance: 5000.0

Account 2:

Account Number: 67890

Account Holder Name: Jane

Balance: 10000.0

Transfer Details:

Amount: 2000.0

Transaction Code: T0001

Sample Output:

Before Transfer:

Account 1: John - 12345 - 5000.0

Account 2: Jane - 67890 - 10000.0

After Transfer:

Account 1: John - 12345 - 3000.0

Account 2: Jane - 67890 - 12000.0

Transaction Details:

Transaction Code: T0001

Amount Transferred: 2000.0

Timestamp: 2023-04-01 10:30:00

Question 51

Create a class Beach with the below attributes:

beachId - int

beachName - String

beachRating - int

beachCost - int

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create class Solution with the main method.

Implement two static methods - findLeastRatingWithName in Solution class.

findLeastRatingWithName method:

This method will take three input parameters - array of Beach objects and string parameter and a int parameter. The method

will return the minimum rating attribute from beach objects whose name is matched with string parameter and whose amount is greater than the int parameter. If no data found of beach objects, then the method should return 0.

Note : All the searches should be case insensitive.

These above mentioned static methods should be called from the main method.

For findLeastRatingWithName method - The main method should print the least rating from the beach array. If return 0, then you should print "No beach found".

Input ->

1001

Puri

3

8600

1002

Digha

5

6200

1003

Digha

3

4000

1004

Digha

4

5500

digha

5000

Output ->

4

QUESTION 48

Define two functions as given below.

1. **validateScore**: This function will validate a given number whether it is less than or equal to 100 and greater than 0.

If these conditions are matched, function will return True, otherwise return False

2. **findValidScores** : This function will take a list of scores (numbers) as input, and return a list of scores from the input

list after validation . Use the method validateScore to validate scores in the input list.

If the input list does not contain

any valid score, function should return None.

Define the main section for the following:

1. Create a list of positive integers by taking user inputs

To create the list, do the following:

a. First read the number of elements (integer) you want to store in the list as input from user

b. Read a number as input and add to the list. Repeat this step for the number of elements to be stored in the list (considered in the first line of input)

2. Call the method findValidScores by passing the list created in point #1 as argument.

3. Print a message "Valid Scores =" (excluding the quotes) followed by the list returned by the method findValidScores in the

same line. If the method returns None, print the message "No valid score found." (excluding the quotes).

Sample Input description:

The first input is the number of elements to be added to the list.

The next set of inputs are the numbers to be added to the list one after another and is repeated for the number of elements

given in the first line of input.

Check the sample input and output below.

Sample Input 1:

For example, if you want to create a list of 5 positive numbers with values 10,20,101,90 and 0

then the input should be read/given as follows:

5
10
20
101
90
0

And the output for the above set of input will be:

Valid Scores = [10, 20, 90]

Sample Input 2:

8
-100
101
0
.80
101
120
-75
0.78

Output:

No valid score found.

QUESTION 51

Create a class Beach with the below attributes:

beachId - int

beachName - String

beachRating - int

beachCost - int

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create class Solution with the main method.

Implement two static methods - **findLeastRatingWithName** in Solution class.

findLeastRatingWithName method:

This method will take three input parameters - array of Beach objects and string parameter and a int parameter. The method

will return the **minimum rating attribute** from beach objects whose name is matched with string parameter and whose amount

is greater than the int parameter. If no data found of beach objects, then the method should return 0.

Note : All the searches should be case insensitive.

These above mentioned static methods should be called from the main method.

For findLeastRatingWithName method - The main method should print the least rating from the beach array. If return 0, then you should print "No beach found".

Input ->

1001

Puri

3

8600

1002

Digha

5

6200

1003

Digha

3

4000

1004

Digha

4

5500

digha

5000

Output ->

4

Question 53

Create a Class Course with below attributes:

CourseName - String

CourseNumber - int

Mode - String

Sharedata - boolean

Write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement static method - **findCourseStatus** in Solution class.

This method will take a **String parameter** along with the **other parameter as array of Course objects**.

The method will **return status** where the **String parameter appears in the side attribute (with case insensitive search)**.

- If the number of the course greater than equal to 1000, **print "High"**, if greater or equal to 500 but less than 1000 then **print "Medium"** and if less than 500 then **print "Low"**.
- If the taken **string not match with the Course name** then print "No course found".

Consider below sample input and output:

Input 1:

Java

1200

Online

False

Python

600

Offline

False

HTML

800

Offline

False

JavaScript

300

Online

True

Python

Output 1:

Medium

=====

Input 2:

Java

1200

Online

False

Python

600

Offline

False

HTML

800

Offline

False

JavaScript

300

Online

True

PHP

Output 2:

No course found

Question 55

Create a class **Resort** with the below attributes:

resortId - int

resortName - String

Category - String

resortPrice - double

resortRating - double

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create class **Solution** with the main method.

Implement two static methods - findAvgPrice in **Solution** class.

findAvgPrice method:

This method will take two input parameters - array of **Resort** objects and a string parameter.

The method will **return the average of the prices** from resort objects whose **Category matched with passed as parameter** and **rating will greater than 4.**

If no resort present found, then the method should return 0.

Note : All the searches should be case insensitive.

These above mentioned static methods should be called from the main method.

For findAvgPrice method - The main method should print the average price as it is if the returned value is greater than 0, or it should print "No such Resort found".

Before calling these static methods in main, use Scanner object to read the values of four resort objects referring attributes in the above mentioned attribute sequence. Next, read the value for category.

Input

1004

ABC2

3 star

5400.00

4.5

1001

DEF

5 star

4781.45

4.0

1002

GHI

3 star

6235.72

4.2

1008

JKL

2 star

7454.00

4.2

3 star

Output

The average price of 3 star:5817